

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming in Java

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Is JVM is machine independent? Explain the use of JVM. How JVM makes Java machine independent.
b) Why is multiple inheritance not supported in Java. Provide a simple code example to support your answer.
2. a) What is reflection? Why do we need reflection? Explain with sample code
b) How can you create your own exception in Java? Explain with program.
3. a) Discuss about the Sandbox Security model with regard to Java Applet.
b) Explain how to display a picture in an applet.
c) Differentiate between GridBag Layout and Grid Layout.
4. a) What is the difference between AWT and Swing package in Java? Write a sample code for creating a closable frame in the front and centre of your desktop.
b) Write a program to draw a Bar Chart of the total number of students appearing exam in Java. Also, provide Bar Chart for number of male and female students.
Hint: Each Bar can be drawn using drawRect() and fillRect() method.
5. a) What is an URL. Write a Java program for reading contents in a given URL.
b) Write a client program which connects with server running at port 5000 and send "I am client" message to server and also print received message from server in console. (Assume TCP/IP communication)
6. a) Differentiate between simple statements and prepared statements. Provide an example to illustrate their differences.
b) Write a simple Java Program to connect database and read data in the table named "Student" which have four fields named "id", "name", "address", "DOB", and "Class". Assuming the "id" field as simple number; display the data of the students with id less than 100.
7. Write short notes on: (Any two)
a) JIT compile
b) Repaint method
c) Inner Class

2

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming in Java

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) How does Java support re-usability? Provide examples.
b) Discuss about the concept of instance variable hiding in Java. Illustrate with a code example.
2. a) Make a class Human with a name and age. Make a class Employee inherit from Human. Add instance variable salary of type double. Supply a method showData() that prints the Employee's name, age, and salary. Make a class Manager inherit from Employee. Supply appropriate showData() methods for all classes. Provide a test program that tests these classes and methods.
b) Why is exception handling essential? Differentiate between checked and unchecked exceptions with relevant examples.
3. a) Discuss about the life cycle of an Applet.
b) Write an Applet program to illustrate the message sharing between HTML and Applets.
c) Differentiate between Panel and Frame.
4. a) Write a program to generate a Frame with two buttons "BLACK" and "BLUE". When a button is clicked it should change the background color of the Frame to respective color.
b) Write a Java Program to draw a 2D rectangle in green color. Next draw the flag of Nepal inside the rectangle. Write a string "MY NEPAL" below the rectangle.
5. a) What is a socket? Differentiate between datagram and stream communication.
b) Write the steps to create a client server program. Provide code snippet for each step.

6. a) What is a ResultSetMetadata. Provide a simple example code to illustrate its usage. 8
b) Discuss about the different types of database drivers. 7
7. Write short notes on: (Any two) 2x5
a) Assignment vs. initialization
b) Closable JFrame
c) Runtime Polymorphism in Java

POKHARA UNIVERSITY

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Semester: Spring

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Attempt all the questions.

1. a) Define java. Write a program to print "Hello Java" in a console. 8
b) Define Polymorphism. How should we implement polymorphism in java? Explain with Example. 7
2. a) What is interface? Define with Example. 8
b) Create a class MyClass in a package MyPack. Import newly created class MyClass from IpmClass. 7
3. a) What is the usage of exception handling? Define with example. 8
b) Create a program to write "Hello World" in a file abc.txt. 7
4. a) Define Applet life cycle. 8
b) What is Frame? Create a frame with following attributes: 7
 Height = 400
 Width = 400
 Title = My Frame
5. a) Create a Frame with one button and one textfield, when user clicks on the button the text entered on textfield should be changed to uppercase. 8
b) Explain Event Handling with various types of event available on java. 7
6. a) Define Socket Programming with example. 8
b) Why JDBC is required in java. Explain the role of JDBC Driver. 7
7. Write short notes on: (Any two) 2
a) Inheritance
b) URL
c) Java Virtual Machine
d) ResultSet

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming in Java

Semester: Fall

Year : 2017
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Time : 3hrs.

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Attempt all the questions.

1. a) Justify the statement, "Java is designed for distributed application".
Explain the different types of Java variables.
b) Explain with example: static block, static variable and static method.
Why main method is always static and public in Java?
2. a) How can you achieve multiple inheritance in java? Explain with a suitable program.
b) What are the main uses of super, finalize and this keyword? Explain how a package is created and accessed while developing an application in Java.
3. a) Write a program to store object of a class Student into a file "student.dat" also read the objects from same file and display the state of objects on the screen. Handle possible exceptions.
b) Differentiate error with exception. Write a program to create your own exception class in java.
4. a) Explain Applet Architecture. How can you convert applet to application? Explain with suitable example.
b) How Event is handled in Java? Write a Java program to create a swing application with 3 buttons representing your favourite colors. When a button is clicked, the background color must change to the corresponding color.
5. a) Differentiate between GridLayout and GridBagLayout. Write a program to show the use of BorderLayout.
b) What is GUI programming? Write a program to draw Nepali flag using graphics.
6. a) What is Java URL processing? Demonstrate URLConnection class

- method with suitable example program.
- b) What are the different types of JDBC statements available? Explain with example. 8
7. Write short notes on: (Any two) 2x5
- a) Socket Programming in Java
 - b) JDBC Driver types
 - c) Reflection in Java

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Attempt all the questions.

1. a) Define Java as Platform Independent Programming Language.
b) What is constructor, Explain with example?
2. a) Write a program to show the usages of various data types in java.
b) Define Inheritance. Write a program which has two classes A and B, where A should act as Parent class and B should inherit from A.
3. a) Define Exception handling with Example.
b) Create a Frame which has three textfield and one button. When user clicks on button it should calculate sum of the value of first and second textfield and display on third textfield.
4. a) What is difference between Interface and Class? Explain with an example.
b) Write a program to read content from file "abc.txt" and store it in "xyz.txt".
5. a) What are applets? Define Applet life cycle with Example.
b) Write a program to send "Message from Pokhara University" from client to server using java socket programming.
6. a) Define JDBC. Write a program to update data on following table.
Table: student
Column:
NAME TYPE
id number
name varchar
age number
b) What is the difference between Frame and Swing?
7. Write short notes on: (Any two)
a) JRE
b) Event handling
c) Result Set

POKHARA UNIVERSITY

Level: Bachelor
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Semester: Fall

Year : 2018
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Attempt all the questions.

1. a) How is interface different from abstract class? Explain the use of interface to achieve multiple inheritances.
b) What are the uses of abstract keyword? Explain with a suitable program.
2. a) Create a class Employee with id, name, post and salary. Create a parameterized constructor to initialize the instance variables. Override the toString() method to display the employee details.
b) What is exception handling? Explain. Write a program to handle Arithmetic Exception.
3. a) What is a dialog box? Explain its types. Write a program to create your own dialog box.
b) What are the different types of streams supported by java? Explain. A data file "emp.txt" contains name, address and salary of 30 employees. Write a program to display only those records who are from "Kathmandu".
4. a) Create a swing GUI that contains a two buttons (add and subtract) and three text fields. When the buttons are clicked sum or difference of values of first two text fields should be displayed in the third text field respectively.
b) Explain the use of URL and URL Connection class with a suitable program.
5. a) Write a program to display all records from the database table. Assume the name of database and table yourself.
b) Differentiate between TCP and UDP. Create a TCP client application that takes input from user and sends it to the server.
6. a) How is applet different from normal java program? List out the steps for

- converting applet into application.
- b) How can you create closable frames in swing and AWT? Write a program to draw a bar chart.
7. Write short notes on: (Any two) 2x5
- a) Reflection API
 - b) History of Java
 - c) JDBC API

POKHARA UNIVERSITY

Level: Bachelor
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Semester: Spring

Year : 2018
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Attempt all the questions.

1. a) What do you mean by architectural-neutral? What are wrapper classes? Explain. 7
b) Mention the scope of all modifiers (private, default, protected and public). Write suitable program to illustrate the concept. 8
2. a) What is method overloading? Can you override a private or static method in Java? Explain with an example. 8
b) How does interface differ from abstract classes? Elaborate using code snippets to justify. 7
3. a) Explain about FileReader and BufferedWriter class. How do you create own exception subclasses? Explain with an example. 8
b) What are the differences between applet and normal java program? Create an applet with the functionalities to play, stop and repeat the audio. 7
4. a) Create a swing GUI that contains a text field and a button. When the button is pressed the content in the text field should be changed into uppercase and background color of text field should be changed. 7
b) Demonstrate various drawing methods. How do you create, load and display image? 8
5. a) Differentiate between TCP and UDP sockets. Explain InetAddress class. 7
b) What are some key classes defined in java to work with datagrams? How do you get a list of IP addresses that are assigned to a network interface? 8
6. a) What is the benefit of using Prepared Statement in java? What is JDBC database connection pool? How to setup in Java? 8

7

- b) A database "testdb" contains a table "employee" with some records having id, name, post, salary. Write a program to update the salary to 50000 whose post is "Manager".

Write short notes on (Any Two):

2×5

- a) Inner Class
- b) C++ Vs Java
- c) Types of JDBC drivers

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2019

Programme: BE

Full Marks: 100

Course: Programming in Java

Pass Marks: 45

Time : 3hrs.

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Attempt all the questions.

1.
 - a) What are the most powerful features of java? Describe how it is a platform neutral language by a diagram.
 - b) Explain multiple inheritance. How is it possible to achieve multiple inheritance in java?
2.
 - a) What is interface? WAP to show interface implementation in java.
 - b) Create a program to write "Test on File Handling" inside a file "abc.txt".
3.
 - a) What is an Applet? Explain life cycle of applet.
 - b) What are the differences between GridLayout and GridBagLayout? Explain FlowLayout.
4.
 - a) Create a swing GUI that contains a button, and two text fields. When the button is clicked the value of first text field should be checked and display "odd number" or "even number" in the second text field.
 - b) Create a graphics application to display "Pokhara University" in blue color with font name: Times New Roman, type:Bold and size:20. Also write HTML to embed the applet.
5.
 - a) Draw a diagram illustrating how client server interaction occurs in UNIX based system, with detailed methods in each step.
 - b) Create a TCP client/server program in which client sends an integer to the server and the server responds to client by sending square of the number sent by client.
6.
 - a) Briefly describe the JDBC-ODBC types of bridge and driver in java.

- b) Write a program to display only those records whose salary is more than 25000 from a table that contains id, name, post and salary of some employees. 8

7. Write short notes on: (Any two) 2×5

- Reflection API
- Events and event classes
- Types of JDBC drivers

POKHARA UNIVERSITY

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Semester: Spring

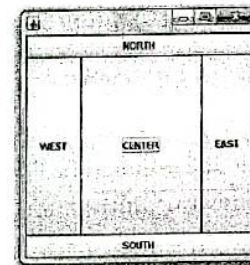
Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- List the advantages of java programming language. Why is it also known as platform independent language? 8
 - In what condition is the use of "this" operator mandatory? Explain with suitable program. 7
- Why Reflection is regarded as a powerful aspect in java? WAP to demonstrate your answer. 8
 - What do you mean by exception? How exception handling is different from error handling? Explain. 7
- State the life cycle of an applet. Can it be changed into application? If yes, justify your answer. 8
 - What is layout manager? State the different types of layout manager in swing. WAP to generate the following output in java using Boxlayout. 7



- Create a swing GUI which contains a text field (to input radius of circle), a label and a button. When the button is clicked area of circle should be calculated and displayed in the label. 8
 - WAP to display the flag of nepal, drawing a graph along with the reasonable coordinates. 7

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Programme: B E

Course: Programming in Java

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

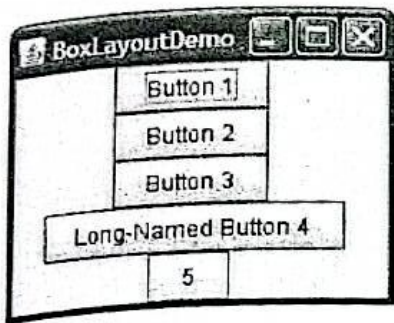
5. a) Explain network programming. Compare TCP and UDP sockets. 7
b) Create a TCP client/server program in which client and server communicate by sending a message to each other. 8
6. a) Draw JDBC architecture and describe different types of JDBC Drivers. 7
b) Discuss the tags used in applet. 8
7. Write short notes on: (Any two) 2×5
 - a) Interface and Inner Classes
 - b) Different event handling mechanisms
 - c) Executing SQL queries from Java

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the Intermediate Working principal of java programming language that makes it runnable on any platform.
b) Why multiple inheritance is not supported in Java. Provide a simple code example to support your answer.
2. a) Why Reflection is regarded as a powerful aspect in java? WAP to demonstrate your answer.
b) Define exception and explain why it is necessary to handle exceptions? WAP to write bytes of data to a file called "exams.txt" and reading it until end of file is obtained, using proper exception handling.
3. a) Define Applet and explain its lifecycle. Create a simple Applet and demonstrate how we can run a Java Applet.
b) Create a Frame with two Text Fields, one of which should show whether the mouse pointer is inside or outside the Frame and the other text field should display the X and Y co-ordinates of the pointer when the user moves the mouse pointer inside the Frame area.
4. a) State the different types of layout manager in swing. WAP to generate the following output in java using Box Layout.



- b) As per a recent survey, among all the Javascript frameworks 40% prefers React while Angular comes second with 30% developers preferring it and 20% of developers prefer Vue while remaining 10% use other frameworks. Create a pie to show case the survey information. 8
5. a) What are the different socket types in java? Draw a diagram with appropriate methods to describe client server interaction in java. 8
- b) Explain the use of URL and URL connection class with a suitable example. 7
6. a) What is JDBC? Draw JDBC architecture and explain JDBC drivers with their advantages and disadvantages. 7
- b) WAP to insert data of any student with schemas (name. age. Semetar. University. Status.) By default. Sqlconnector is to be used (if comfortable with any database connector. free to describe and use accordingly) 8
7. Write short notes on: (Any two) 2×5
- Wrapper class.
 - Life cycle of an applet.
 - TCP and UDP socket.

POKHARA UNIVERSITY

Level: Bachelor

Semester – Spring

Year: 2020

Program: BE

Full Marks: 70

Course: Programming in JAVA

Pass Marks: 31.5

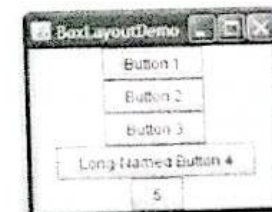
Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

Group - A: (5×10=50)

- Q. N. 1 Illustrate the scope of static and local variable with an example. Differentiate between applet and Java application 10
- OR**
- The final keyword in Java can be applied to fields, parameters, methods and classes.
For each of these four uses, explain what the effect of the final keyword is. 10
- Q. N. 2 What are the types of I/O streams.
Write a program in java using FileWriter to write "Pokhara University is awesome for programming" in a new file. 10
- Q. N. 3 Why Reflection is a powerful tool used in software testing and debugging?
Elaborate with a small example. 10
- Q. N. 4 Define JDBC architecture and differentiate type 1 versus type 4 JDBC driver with merits and demerits.
WAP to connect a java program to database successfully. 10
- Q. N. 5 State the different types of layout manager in swing. WAP to generate the following output in java using BoxLayout. 10



Level: Bachelor
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Course: Programming in Java

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3 hrs

Group - B: (1×20=20)

- Q. N. 6 Write a program to create a GUI based application that stores the user input in a database. 20



The program should have following requirements

- Delegate separate class for Even handling, GUI display and database operations.
- When Insert button is pressed the information entered in the fields should be saved in a database.
- The Insert button should be disabled unless all the fields are filled
- After every database entry the user should be informed using dialog box whether the operation has been successful or not
- The Exit button should close the application

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks

Attempt all the questions.

- Write a Java program to get 10 array elements from user and put even and odd elements in separate array. Explain `public static void main (String [] args){ }`
 - Differentiate between Is-A and Has-A relationship. Is multiple inheritance is possible in Java? Explain.
- Write a program in Java to access a constructor, a private field and a private method using Reflection API.
 - Define exception and explain why it is necessary to handle exception? WAP to write bytes of data to a file called "Java_Exam. jsp" and reading it until end of file is obtained, using proper exception handling.
- Define Applets. Draw digital clock using applets.
 - How event is handled in Java? Write a program to draw pie chart using AWT.
- Create an applet that lets user to draw in a canvas and the save the drawing as an image file
 - In a class of 100 students 20 of them prefers java while 35 students use php, 30 students use python and remaining uses Go Lang as their preferred backend language. Illustrate this stat using a bar diagram
- Define Java Socket Programming. Write java program to create own server and client.
 - Create a multithreaded TCP server that can take message from multiple user and reply back to them individually.
- What is JDBC. Explain JDBC drivers
 - A database "pu" contains a table "Employee". Write a program that asks user to enter a name, then query the database and display total number of employees with that name.
- Write short notes on: (Any two)
 - How Java becomes Platform Independent?
 - Differentiate between Grid Layout and GridBag Layout.
 - Differentiate between Connection-oriented and Connection-less programming.

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2022

Programme: BE

Full Marks: 100

Course: Programming in Java

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write a Java program to get 10 array elements from user and delete x^{th} position element from array. Explain
`public static void main(String[] args){ }` 8
b) What are the pros and cons of Reflection API? Write a sample program that does the reflective operation of accessing a private method of a class. 7
2. a) Differentiate between process and thread. How multithreading is performed in java? Illustrate with example. 8
b) Define exception and explain why it is necessary to handle exceptions? WAP to write bytes of data to a file called "javaExam.jsp" and reading it until end of file is obtained, using proper exception handling. 7
3. a) What is an applet? Create an applet that takes a value from the HTML in which it is embedded and displays it. 8
b) How to close AWT Window in Java? Explain all methods with example. 7
4. a) Create a popup menu attached to frame, if you click within the area of frame popup menu will be shown. Similarly clicked on menu item show message "..... item is selected" on label. 8
b) Create a frame with a red circular object in the center. Also add four buttons named up, down, right and left. When a button is pressed the ball should move in the direction corresponding to the name of the button. 7
5. a) Write a menu based program that does the following tasks: 8
 - i. Print out all the IP addresses associated with Netflix.com
 - ii. Display the localhost address
 - iii. Print the loopback address
 - iv. Exit

- b) Create a TCP client/server program in which multiple user can send and receive message from each other. 7
5. a) What is JDBC? Explain JDBC drivers types. 8
- b) A database db_std contains a table tbl_std having field roll, name, faculty and mark. Write a program that does achieves the following: 7
- i. Print the total number of students
 - ii. Find the average mark
- Write short notes on: (Any two) 2×5
- a) Custom Exception
 - b) Wrapper class
 - c) TCP Vs UDP

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain the Java architecture and significance of class path. 8
Differentiate between error and exception.
- b) Define Package. Explain the types of access modifiers in Java. 7
2. a) Define interface. Write Java program that override methods (getUserValue() and displayUserDetail()), where getUserValue() is defined to get user detail(name, address, age) and displayUserDetail() to display user detail. 8
- b) Define subclass and super class. How can we achieve multiple inheritance in Java? Explain. 7

OR

Differentiate between Java abstract class and interface with a sample program

3. a) Write a program to create a Frame that has two TextField components, one Label, and a Button. When the user clicks on the button, calculate the sum of the values entered in the first and second TextField and display the result on the third Label. 8
- b) What are the pros and cons of JavaFX. Explain any two JavaFX layout managers. 7
4. a) Create a TCP client server application where the client sends a string and the server responds by echoing the same string in upper case. 8
- b) What is JavaMail API? Explain type of protocol used in JavaMail API. 7

OR

Define RMI stubs and skeleton. Create an RMI application where a client can remotely invoke a method that sends the sum of any two given integers.

5. a) A table `tbl_student` consists of `id`, `name` and `program`. Write a program that asks users to enter a program name and displays all the students records enrolled in that program. 8
- b) Differentiate between GET and POST methods. Write java program that depict Sessions and Cookies. 7
6. a) Explain life cycle methods of servlet. Write a servlet program that accept input string from user through form and then check if it is palindrome or not. 8
- b) Why do we need design patterns? Explain Singleton Pattern and Factory Method Pattern in detail. 7
7. Write short notes on: (Any two) 2×5
- a) SQL Injection
- b) ORM
- c) CORBA

POKHARA UNIVERSITY

Level: Bachelor

Programme: BE

Course: Advanced Programming with Java

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the role of the JVM in Java Architecture? Describe the steps to compile and run a Java program. 7
- b) Write a multithreaded program in Java to print even and odd numbers alternatively. 8

OR

What are design patterns? Compare the Singleton pattern with the Factory pattern.

2. a) Define Interface. As we know, there is no room for multiple inheritance using classes. How do you achieve multiple inheritance in java. Explain with Example. 8
- b) What is a constructor in Java? Explain the different types of constructors with examples. 7
3. a) Explain the event handling mechanism in Java. Write a program that uses ActionListener to handle button clicks. 7

OR

Define stage and scene in JavaFX. Differentiate between Swing and JavaFx.

- b) Compare the different layout managers in Java Swing with examples for each. 8
4. a) Develop RMI application where client input two numbers if both numbers are even then result $(a+b)^2$ otherwise $(a+b)^3$. 8
- b) Write a Java program to implement a client-server application using TCP sockets. Explain the flow of communication. 7
5. a) What are the main components of JDBC architecture? Explain the role of the DriverManager and Connection interface. 7

- b) Write a Java program to establish a connection to a database using JDBC. Demonstrate the use of *Statement* and *PreparedStatement* objects with examples. 8
6. a) What are the differences between static and dynamic web applications? How does a servlet fit into a dynamic web application? 7
- b) Discuss the life cycle of a servlet in detail, including the methods *init()*, *service()*, and *destroy()*. 8
7. Write short notes on: (Any two) 2×5
- a) Handling GET/POST Requests
- b) Hibernate
- c) Advantages of MVC

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Advance Programming with Java

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define ClassPath? Write a program with custom exception handler that handle arithmetic exception. 7
b) Explain the architecture of Java, including the role of the JVM, JRE, and JDK. 8
- OR**
- Define collection framework in java. Explain the different components of java collection framework.
2. a) What are the implications in Java when defining a variable, method, and class as final? Explain the various access modifiers in Java and their significance. 8
b) Define abstraction. Write Java program that override methods (getUserValue() and displayUserDetail()). Where getUserValue() is defined to get user detail(name, address, age) and displayUserDetail() to display user detail. 7
3. a) Pros and Cons of JavaFx. Create GUI application with two buttons "RED" and "Green". When clicking "RED" button change background of "GREEN" button to red and clicking "GREEN" button change background of "RED" button to green. Also, console button clicked message. 8
b) Define stage and scene in JavaFX. Explain any two JavaFX layout managers. 7
4. a) Explain RMI architecture. Develop RMI application where client request to identify given number is Odd or Even. 8
b) Write a TCP client server program, where a client sends an integer value and server responds by sending the squared value of that integer. 7

5. a) Differentiate between Statement and PreparedStatement. Consider db_college and tbl_student as database name and table respectively where a tbl_student has columns name, faculty and batch. Write Console application with CREATE, READ, UPDATE and DELETE queries only using Prepared Statement. 8
- b) What is a JDBC driver? Explain the different JDBC drive types and can figuration 7

OR

- Write all steps to connect Java Application with any Database. Insert student detail (roll,name,address,age) into table "student_tbl"
6. a) Describe the life cycle of a servlet. Explain the use of doGet() and doPost() methods of the Servlet with an example. 7
- b) Create a Servlet application to showcase the concept of session and cookies 8
7. Write short notes on: (Any two) 2×5
- a) Design patterns
- b) Over view of ORM
- c) RMI vs. CORBA

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define the term Architectural Neutral. Write down the code to handle user defined exception for LowBalance when balance is less than 10 rupees on your mobile. 7
- b) Define multithreading. Explain the techniques to create thread using anonymous class. 8

OR

Write down the advantage of ORM over JDBC. Explain Singleton Pattern and Factory method pattern in detail.

2. a) Explain the concept of inheritance in Java. What are the benefits of using inheritance, and how does it promote code reusability and polymorphism? Provide an example to support your explanation. 8
- b) Differentiate between abstract class and interface? Explain final classes and methods. 7
3. a) Write down the JavaFx code to design a login page as shown in the figure below when user click on login button test the username is "PU" and password is "PU@2025Spring" or not. 8



- b) Why do we need layout management? Explain GridBag layout with suitable example. 7
4. a) What is the significance of stub and skeleton in RMI? Create a RMI application such that a client sends an integer number to the server and the server return the factorial value of that integer. 8
- b) Create a TCP client server application where the client sends a string and the server responds by echoing the same string in upper case. 7
5. a) A table student consists of id, name and program. Write a program that asks users to enter a program name and displays the total number of students enrolled in that program. 7

OR

- What is SQL Injection Attack? Write sample program using prepared statement to prevent SQL Injection Attack. 1.
- b) What is package in java? How can you create user-defined package in java? Explain with example. 8
6. a) Differentiate between servlet and JSP. Create a HTML file with two text fields to First name and Last name. Then create a JSP file that reads values from the HTML form and display full name. 8 2.
- b) What is cookie? How can you store and read cookies by using servlet. Explain with proper example. 7
7. Write short notes on: (Any two) 2×5
- a) Hibernate ORM
- b) CORBA
- c) SQL Escapes
- 3.